# EEE104 – Digital Electronics (I)

# Lecture 14

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Emerging Electronics and Energy Technologies Research Centre (EEET)

Office: SC242

Office hour: 1:00-5:00 pm, Wednesday

# In This Session

Flip-Flops and Related Devices

- Latches
- Edge-Triggered Flip-Flops

  S-R, D and J-K flip-flops;
  edge-triggered operation;
  asynchronous preset and clear

- Flip-Flop Applications

  Parallel data storage, frequency division and counting

# Latches and Flip-Flops

Flip-flops are edge-triggered or edge-sensitive whereas gated latches are level-sensitive.

| Device | When can Q update? | Otherwise |
|---|---|---|
| Gated D latch | While EN is HIGH | Hold |
| Edge-triggered D flip-flop | At the active CLK edge | Hold |

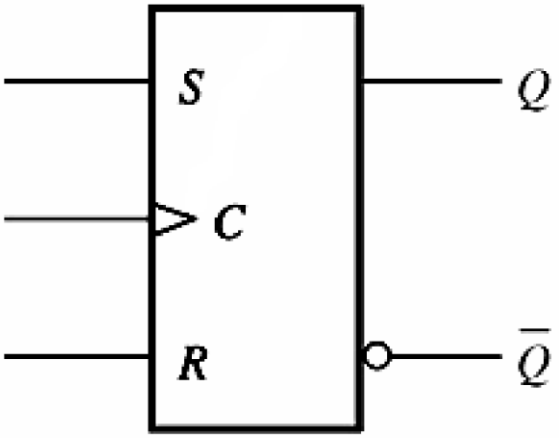
For normal operation with asynchronous inputs inactive.

**EN HIGH is a time interval. A clock edge is a transition.**

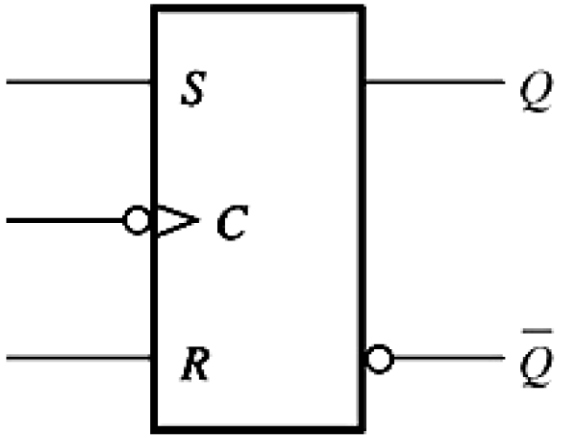
# Edge-Triggered Flip-Flops

An **edge-triggered flip-flop** changes state either at the **positive edge (rising edge)** or at the **negative edge (falling edge)** of the clock pulse.

It is identified by a small triangle at the **clock (C) input**.




Rising-edge triggered

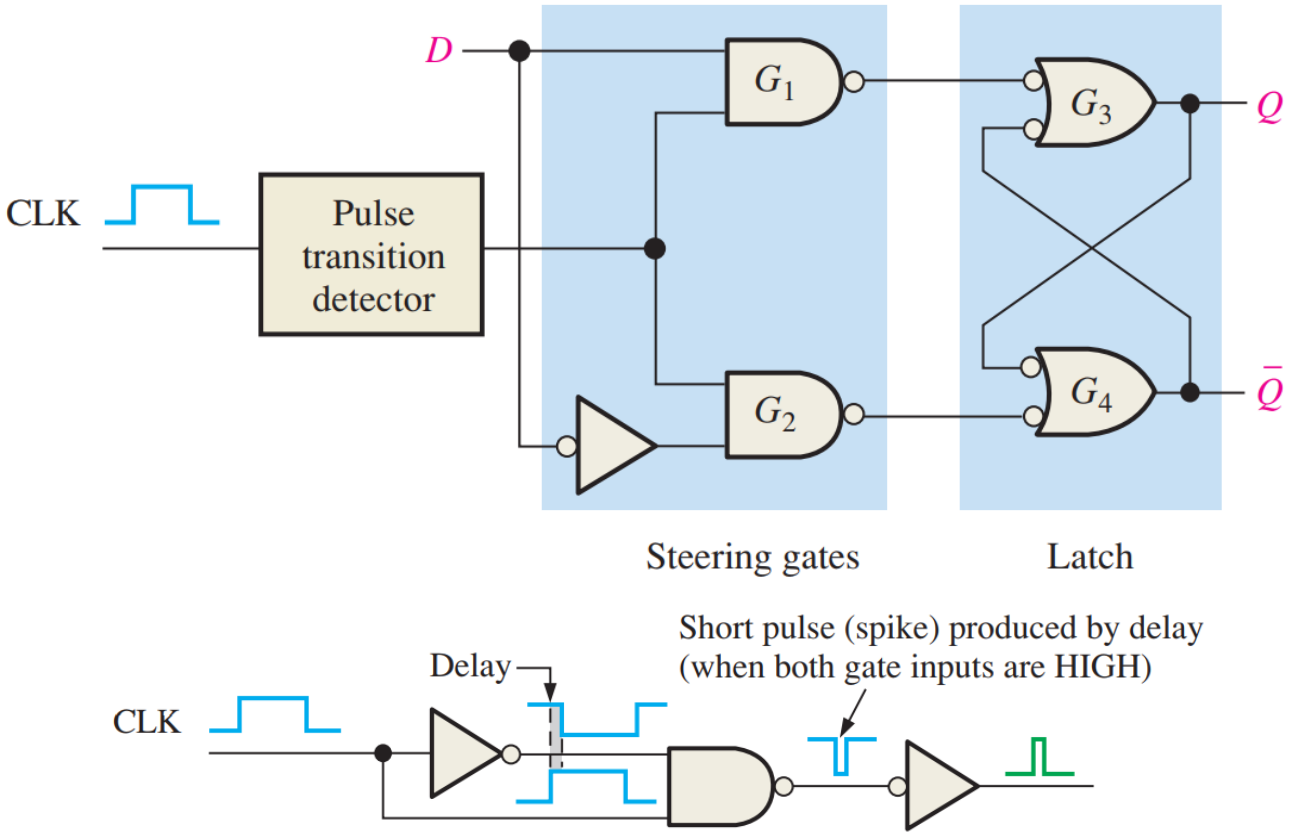



Falling edge triggered

**It is sensitive to its inputs only at this transition of the clock.**

# Edge-Triggered Flip-Flops

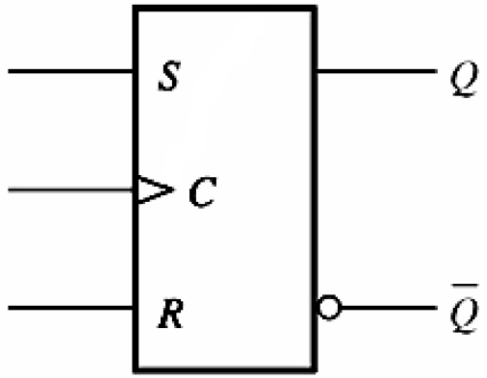
## A Method of Edge-Triggering




This circuit produces a very short-duration spike on the positive-going transition of the clock pulse.

**A steady HIGH clock does not keep the latch enabled.**

# Edge-Triggered S-R Flip-Flops




**Read S and R at the rising edge.**

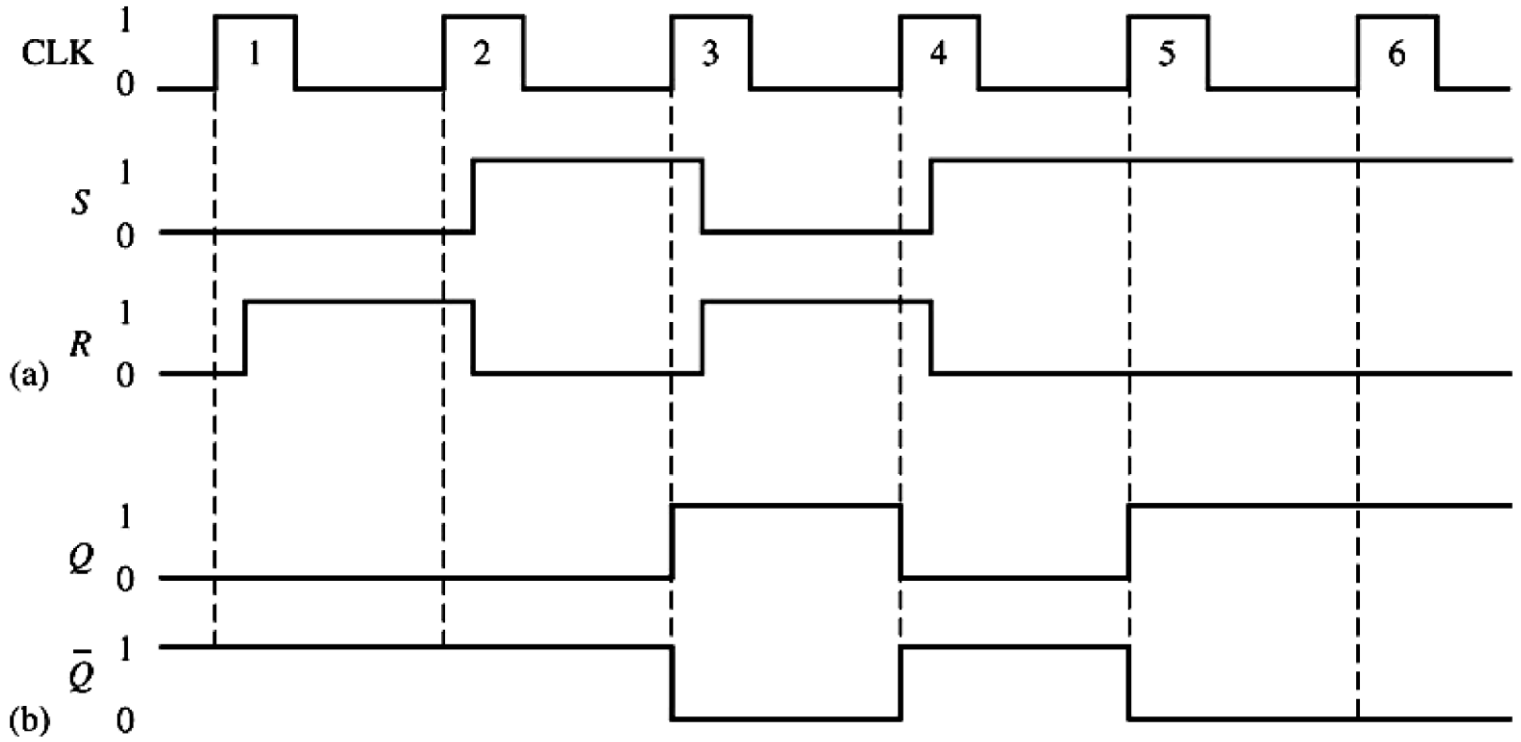
$Q_0$ is the state before the edge.

| S | R | CLK | Q | $\overline{Q}$ | Comments |
|---|---|---|---|---|---|
| 0 | 0 | X | $Q_0$ | $\overline{Q}_0$ | No change |
| 0 | 1 | ↑ | 0 | 1 | RESET |
| 1 | 0 | ↑ | 1 | 0 | SET |
| 1 | 1 | ↑ | ? | ? | Invalid |

↑ means LOW-to-HIGH transition. X means any clock level.
Between active edges, the stored state is held.

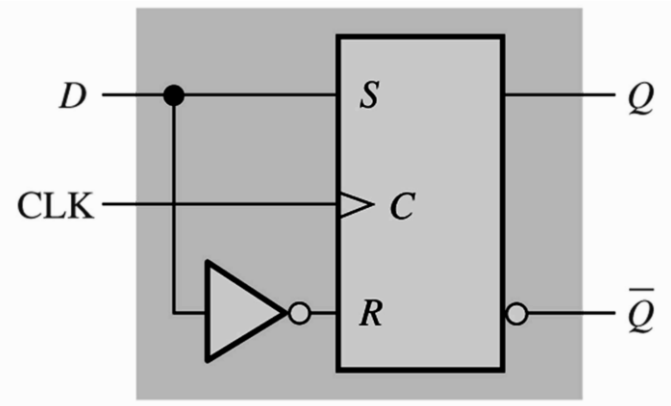
# Edge-Triggered S-R Flip-Flops

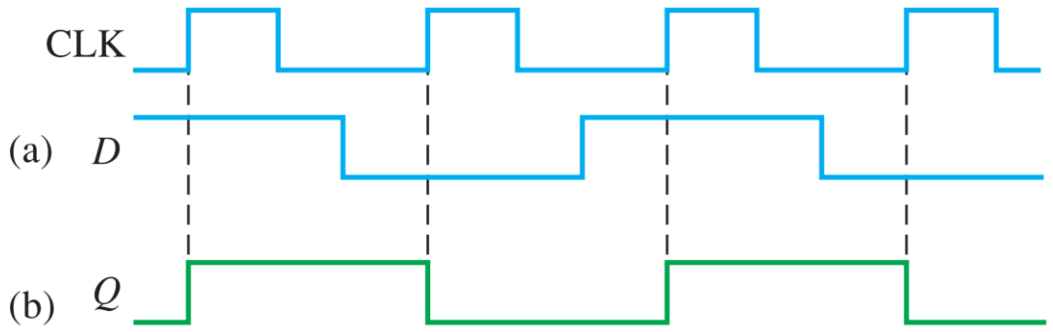
An Example: initial Q = 0




**Read S and R at each rising edge, then hold the result.**

# Edge-Triggered D Flip-Flops



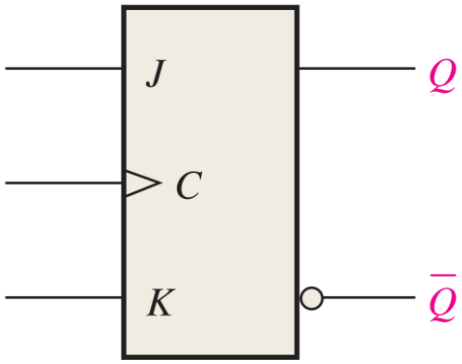




| D | CLK | Q | $\overline{Q}$ | Comments |
|---|---|---|---|---|
| 1 | ↑ | 1 | 0 | SET (stores a 1) |
| 0 | ↑ | 0 | 1 | RESET (stores a 0) |

↑ = clock transition LOW to HIGH

**Q follows D at the triggering edge of the clock.**

# Edge-Triggered J-K Flip-Flops



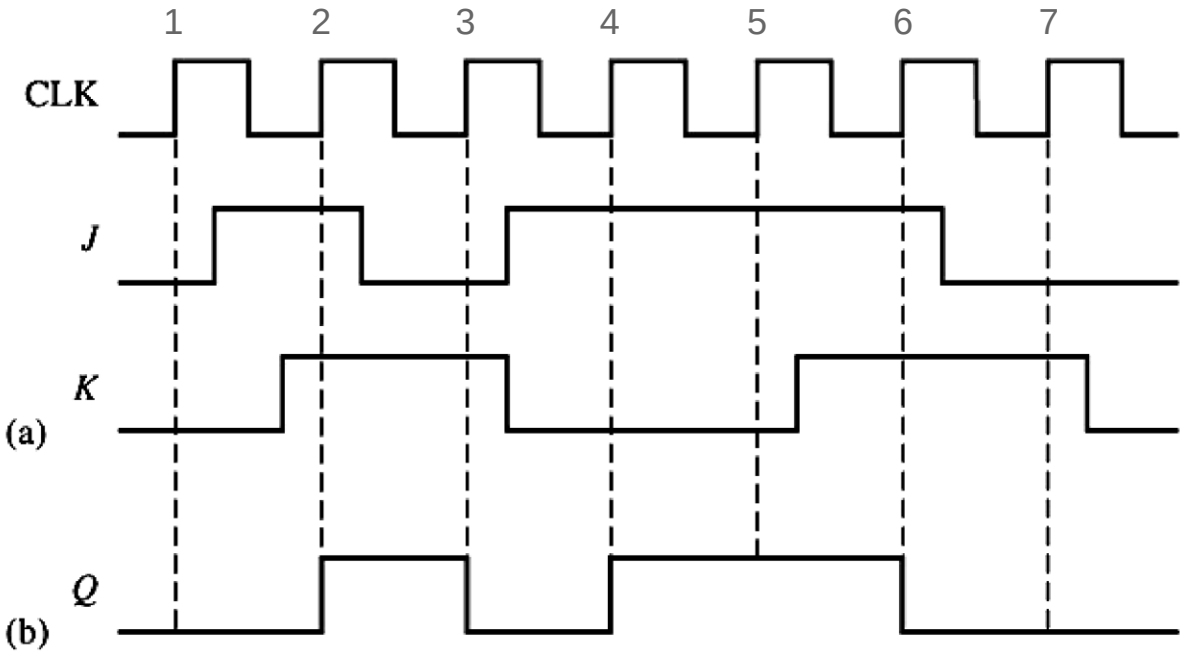

The J-K flip-flop is similar to the S-R flip-flop but has no invalid J-K input combination.

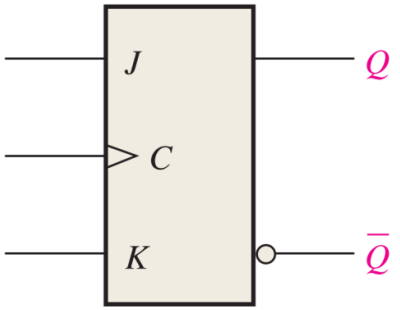
When J = 1 and K = 1, the output will be **toggled** at the rising edge of the clock.

| J | K | CLK | Q | $\overline{Q}$ | Comments |
|---|---|---|---|---|---|
| 0 | 0 | ↑ | $Q_0$ | $\overline{Q}_0$ | No change |
| 0 | 1 | ↑ | 0 | 1 | RESET |
| 1 | 0 | ↑ | 1 | 0 | SET |
| 1 | 1 | ↑ | $\overline{Q}_0$ | $Q_0$ | Toggle |

# Edge-Triggered J-K Flip-Flops

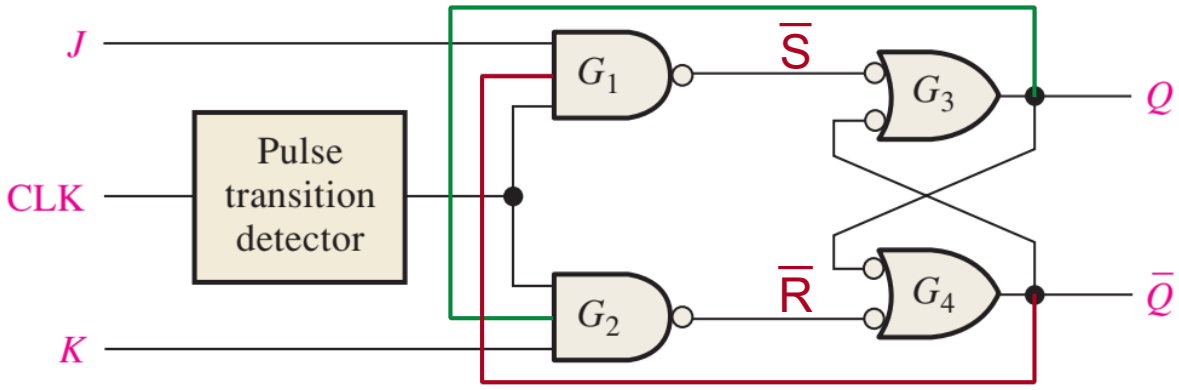
An Example: initial Q = 0




| J | K | CLK | Q | $\overline{Q}$ | Comments |
|---|---|---|---|---|---|
| 0 | 0 | ↑ | $Q_0$ | $\overline{Q}_0$ | No change |
| 0 | 1 | ↑ | 0 | 1 | RESET |
| 1 | 0 | ↑ | 1 | 0 | SET |
| 1 | 1 | ↑ | $\overline{Q}_0$ | $Q_0$ | Toggle |

# Edge-Triggered J-K Flip-Flops




The difference with its S-R counterpart:

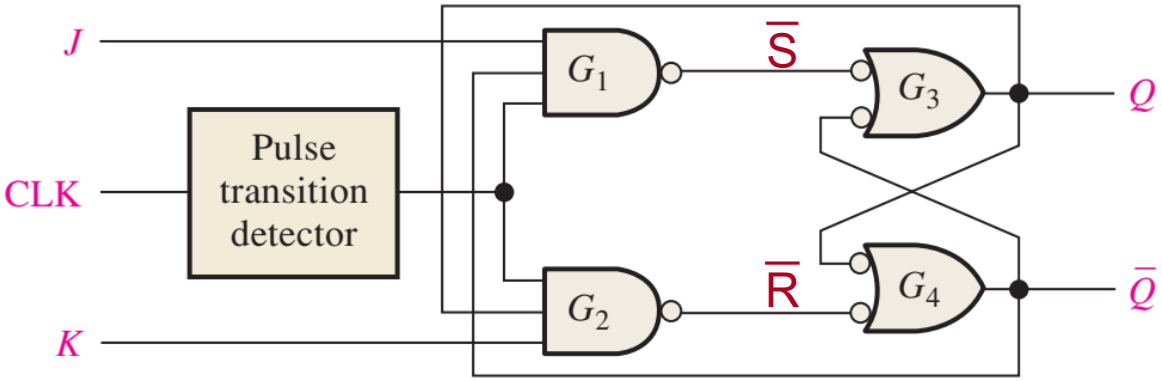
The Q output is connected back to the input of gate $G_2$.

The $\overline{Q}$ output is connected back to the input of gate $G_1$.

$\overline{S}$ and $\overline{R}$ are internal active-LOW latch inputs.

**Q is the state before the edge. Q* is the state after it.**

# Edge-Triggered J-K Flip-Flops




When J = 1 and K = 0, $\overline{R}$ = 1 and $\overline{S}$ = Q at triggering edges of CLK.

$$\overline{S} = \overline{1 \cdot \overline{Q} \cdot 1} = Q$$

**SET operation, Q*=1**

If Q = 0, it will be SET, Q* = 1
If Q = 1, no change, Q* = 1.

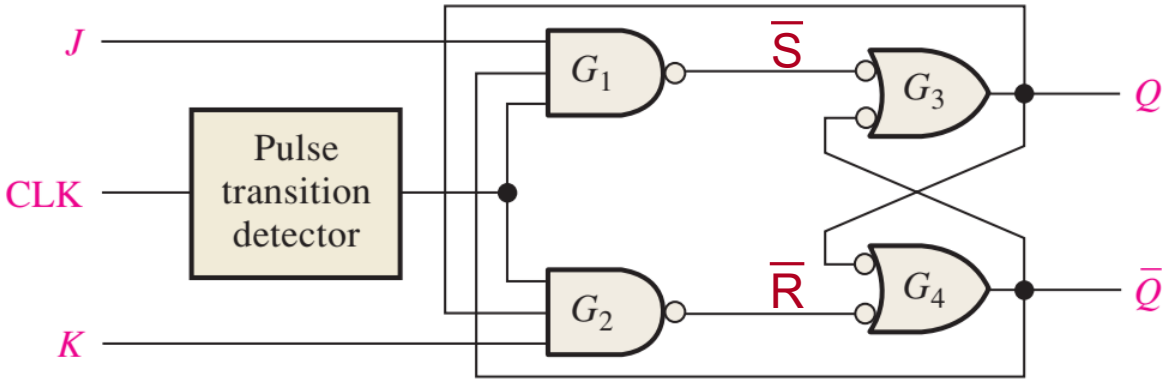
When J = 0 and K = 1, $\overline{S}$ = 1 and $\overline{R}$ = $\overline{Q}$ at triggering edges of CLK.

$$\overline{R} = \overline{1 \cdot Q \cdot 1} = \overline{Q}$$

**RESET operation, Q*=0**

If Q = 1, it will be RESET, Q* = 0
If Q = 0, no change, Q* = 0.

# Edge-Triggered J-K Flip-Flops



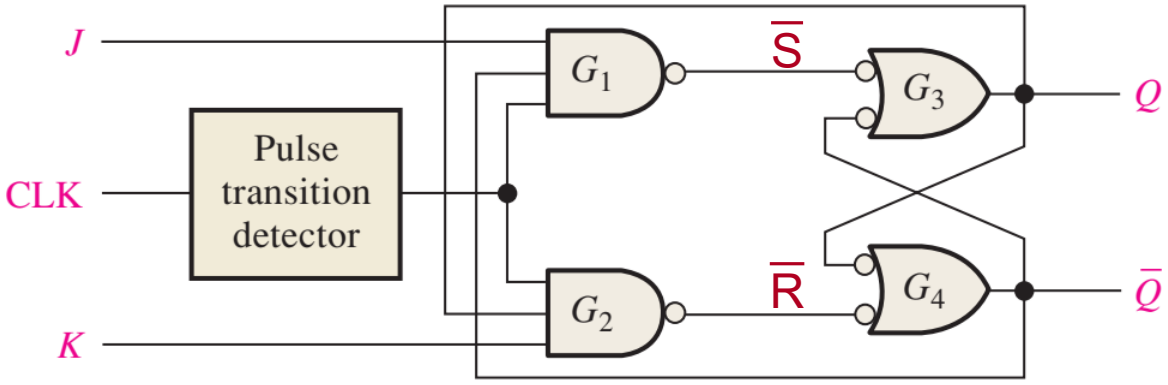

How does it work?

A LOW on both J and K results in a no-change condition.

When J = 0 and K = 0, $\overline{S}$= $\overline{R}$ =1. Q* will not change.

Q* = Q **No change**

# Edge-Triggered J-K Flip-Flops




When J = K = 1

Q functions as the RESET signal. $\overline{S} = \overline{1 \cdot \overline{Q} \cdot 1} = Q$

$\overline{Q}$ functions as the SET signal $\overline{R} = \overline{1 \cdot Q \cdot 1} = \overline{Q}$

If Q = 1, $\overline{S}$ = 1, $\overline{R}$ = 0, Q* = 0.

If Q = 0, $\overline{S}$ = 0, $\overline{R}$ = 1, Q* = 1.

**$Q^* = \overline{Q}$** **Toggle**

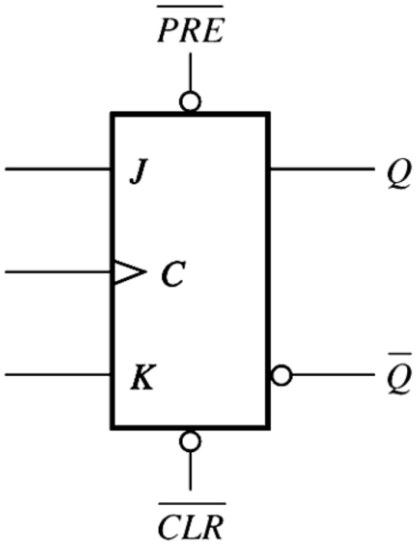
**In the toggle mode, a J-K flip-flop changes state on every clock pulse.**

# Edge-Triggered J-K Flip-Flops

Asynchronous Preset and Clear Inputs

**Synchronous inputs** affect the state of the flip-flop only on the triggering edge of the clock.

**Asynchronous inputs** affect the state of the flip-flop independent of the clock.



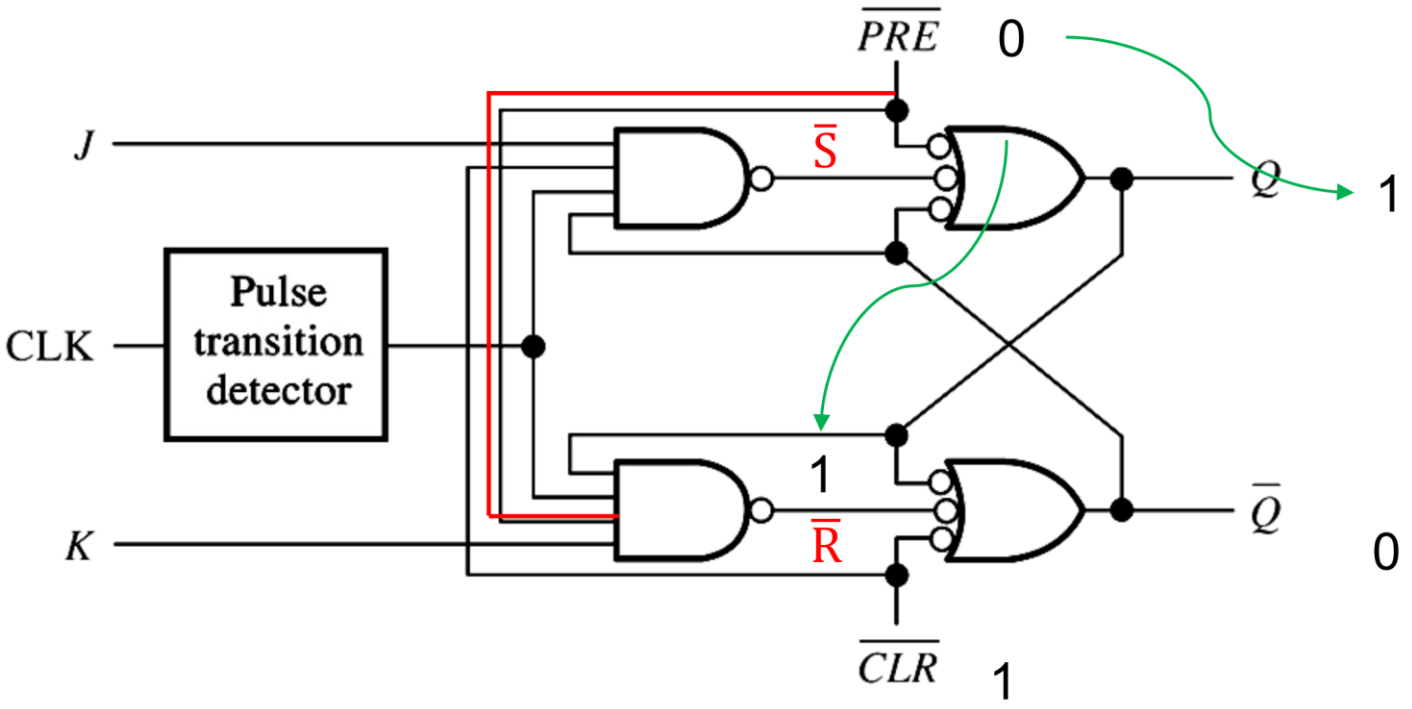

An active level on the preset input will set the flip-flop, $\overline{PRE}$=0

An active level on the clear input will reset it, $\overline{CLR}$=0

**$\overline{PRE}$ = $\overline{CLR}$ = 1 for synchronous operation.**

# Edge-Triggered J-K Flip-Flops

Asynchronous Preset



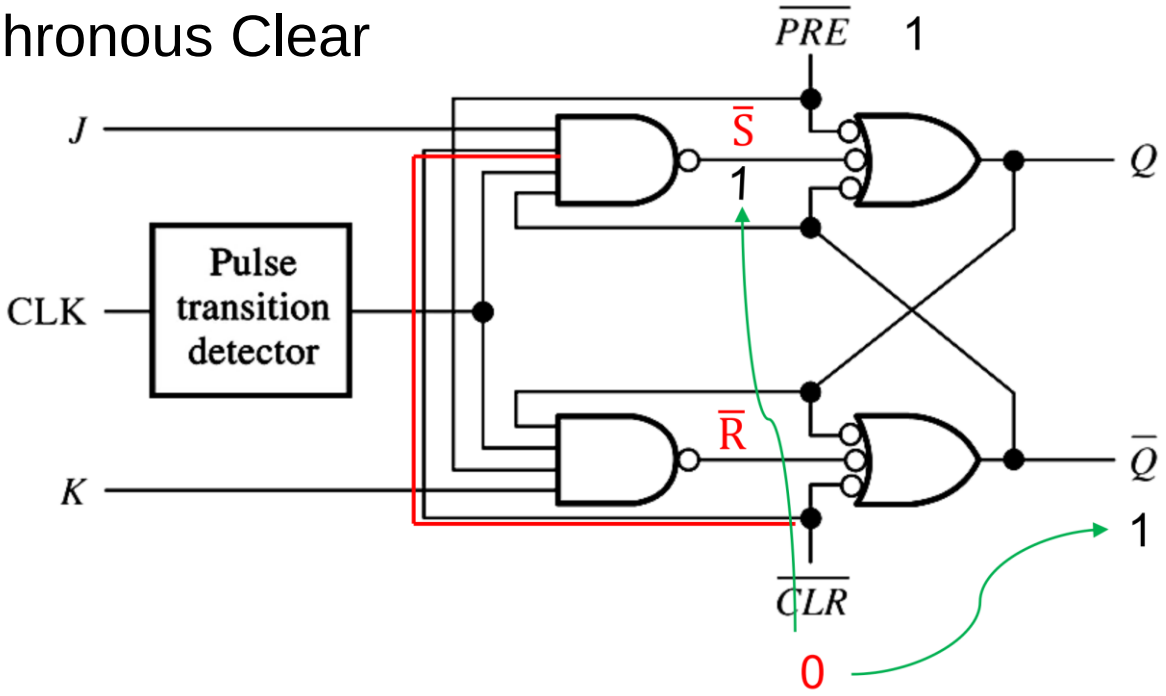

An active preset input makes the Q output HIGH (SET).

When $\overline{PRE}$ = 0 and $\overline{CLR}$ = 1, $\overline{R}$ = 1. Q will be SET.

**Preset acts without waiting for a clock edge.**

# Edge-Triggered J-K Flip-Flops

Asynchronous Clear



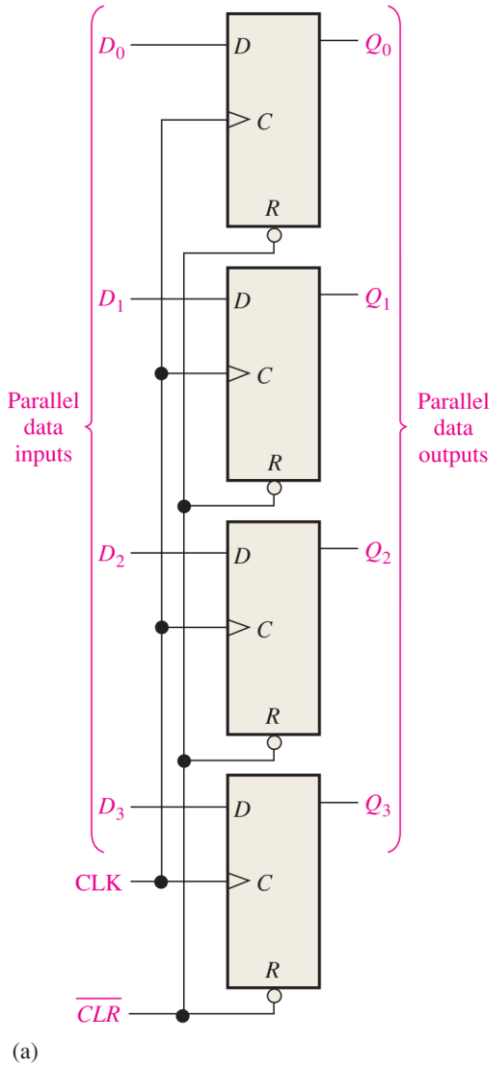

**Q = 0**

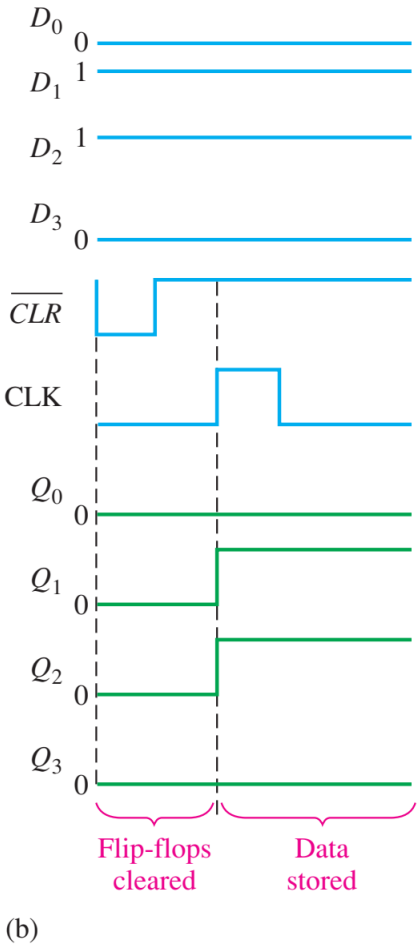
When $\overline{CLR} = 0$ and $\overline{PRE} = 1$, $\overline{S} = 1$. Q will be RESET.

| $\overline{PRE}$ | $\overline{CLR}$ | Normal-use rule |
|---|---|---|
| 1 | 1 | Normal synchronous operation |
| 0 | 0 | Invalid: do not apply together |

# Flip-Flop Applications

## Parallel Data Storage




The clock inputs of the flip-flops are connected together, so that each flip-flop is triggered by the same clock pulse.

$D_3\ D_2\ D_1\ D_0 = 0\ 1\ 1\ 0$

**Clear first.**
**Then store all four bits at one clock edge.**

# Practice 1: Parallel Data Storage

(a) How many D flip-flops are needed to store 4 bits in parallel?

(b) With a common CLK, how many bits are stored simultaneously by one clock pulse?

# Flip-Flop Applications

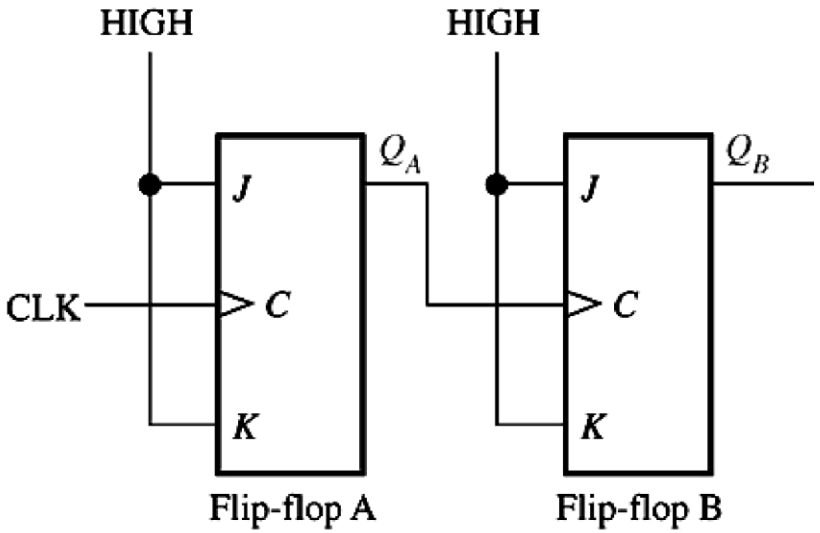
Frequency Division

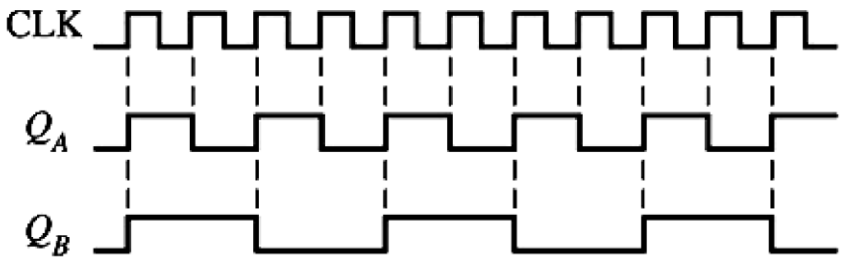
By connecting flip-flops in this way, a frequency division of $2^n$ is achieved, where n is the number of flip-flops.

$f_A = f_{CLK} / 2$

$f_B = f_{CLK} / 4$

J = K = 1
Two toggles make one complete output cycle.



# Practice 2: Frequency Division

**J = K = 1; input CLK = 12 kHz.**

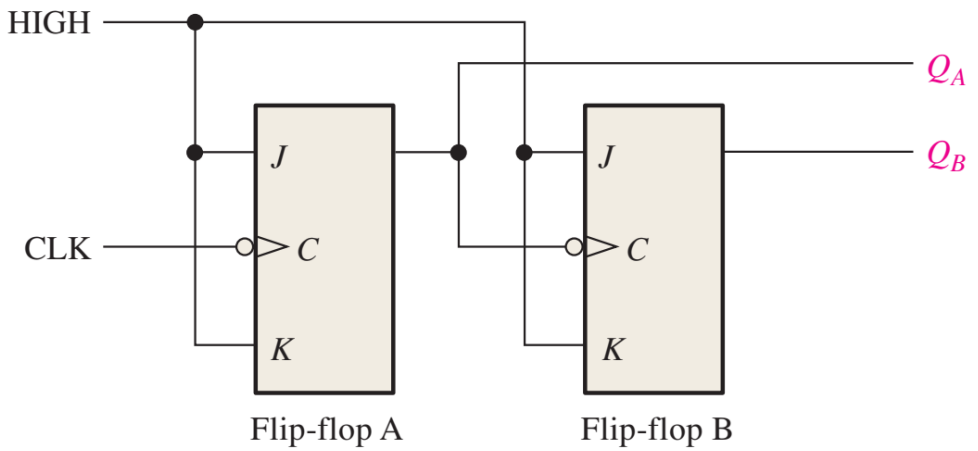
(a) What is the Q frequency of one flip-flop?

(b) A second identical flip-flop is clocked by Q.
What is its output frequency?

# Flip-Flop Applications

Counting

Negative edge-triggered J-K flip-flops are used.




Both flip-flops are initially RESET.

If we take $Q_A$ as the least significant bit, a 2-bit sequence is produced as the flip-flops are clocked.

**Read the count as $Q_B$ $Q_A$.**

**00, 01, 10, 11, 00, ...**

